

Research Craft Notes

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1 Responding to reviews

1.1 What the evidence says

Most reviewers do not move. Across ICLR 2024 and 2025, roughly 75 to 81 percent of scores stayed where they were after the author response, 17 to 23 percent went up, and about 1 percent went down (Farhat et al., *Insights from the ICLR Peer Review and Rebuttal Process*, 2025; *Rebuttals Move Peer-Review Scores*, 2026). The asymmetry matters: writing a response is close to free in expected value, since the downside is rare.

Engagement is the largest single lever. Reviews that received no author reply rose 2.7 percent of the time; reviews that received at least one reply rose 27.0 percent of the time. Within exchanges where authors engaged substantively, the rate reached 45.6 percent, though selection inflates that figure.

Movement is worth chasing because of what it converts into. Papers whose scores rose after the response were accepted 55.7 to 57.6 percent of the time. Papers whose scores did not move were accepted 7.8 to 12.4 percent of the time. For a borderline paper the response is not decoration.

The ceiling is set before you write. Initial score position dominates every feature of the exchange (odds ratio 0.141 per standard deviation). A response cannot rescue a paper the panel has already placed far from the line; it can move one that sits near it.

One finding should shape how a response is drafted. The features that predict movement are mostly the *absence* of failure rather than the presence of brilliance: what a model detects in successful exchanges is that the authors did not make the usual mistakes. Write to avoid the known failure modes first, and only then to persuade.

1.2 Validated features

From the cross-validated feature set that survived Bonferroni correction:

Raises the score	OR	
Baseline completion	1.64	running or accounting for the baseline that was asked for committing to specific presentation changes
Non-defensive tone	1.51	
Writing fixes	1.27	
Theoretical grounding	1.18	
Lowers the score	OR	
Inadequate explanation	0.31	
Low-effort response	0.38	
Defensive tone		
Weak novelty justification		

Four further behaviors are reported as negatives without odds ratios: superficial treatment of a concern, missing baselines, vague promises about future work, and pointing at a section number instead of explaining the content.

That last pair deserves care. Two of the strongest positives, baseline completion and writing fixes, are things a response can only promise when the window is short. The distinction the evidence

draws is between a promise that is specific and checkable and one that is not. “We will improve the evaluation” is a vague promise. “We will run the fourth ablation cell, Factor with no opponent, and report the complete two-by-two” is a commitment someone can hold you to.

1.3 Structure

Answer first, explain second. Rule 7 of Noble’s *Ten Simple Rules for Writing a Response to Reviewers* (PLOS Computational Biology, 2017): give the direct answer, yes or no, before the background. A reviewer scanning for whether you addressed their point should not have to read a paragraph to find out.

Headings should summarize the response, not name the weakness. Maarten Sap’s rebuttal notes make this concrete: prefer “Clarifying the statistical significance” over “Are findings significant?”. A heading that names the weakness makes the reviewer reread their complaint; a heading that names the answer tells them the state of play.

Group shared concerns. When two reviewers raise the same objection, answer it once in a consolidated section rather than twice in two places. This also signals that the reviews were read together.

Keep the reviewer’s own phrasing. Echoing a distinctive phrase from the review lets a reviewer find the response to their point without hunting. This matters more when the response has no line numbers to cite.

Order by what can move. Nothing in the literature says to answer reviewers in numerical order. Weight the words toward the scores with room to move and toward the objections that go to validity rather than presentation.

1.4 Tone

Accept the blame. Rule 3: when a reviewer misunderstood something, treat it as evidence that the text was unclear. This is not only diplomacy. A reviewer who misread a passage is a sample of one from the population of future readers, and the misreading is data about the paper.

Third person, not second. “We agree with the reviewer”, not “we agree with you”. The response is read by the whole panel and by the chairs, not only by its addressee.

Clarify rather than correct. “We will clarify that E counts episodes per iteration” lands better than “the reviewer has misread Table VIII”, and carries the same information.

Write it twice. Rule 10: draft once to work out what you think and to get the irritation out, then write the version that gets sent. The two documents have different jobs and should not be the same file.

Do not validate. Sentences of the form “this is a fair point”, “the concern is valid”, “the reviewer is right that” grant permission the reviewer did not ask for and quietly reverse who is judging whom. Acknowledge by stating the fact instead: not “B is right that the sentence does not establish parity” but “the sentence claims parity without establishing it”.

1.5 Handling disagreement

Do what was asked when you can. Rule 8: even an experiment you consider uninformative is usually cheaper to run than to argue against, and running it removes the objection rather than contesting it.

Cite the other reviewers. Sap suggests supporting a point with what another reviewer said. When one reviewer reads a design as thin and another credits it as a strength, the second review is admissible evidence and costs no words to invoke.

Never argue bias in the response itself. Both Noble and Sap route that to the editor or area chair separately. In the response it reads as a complaint and spends credibility.

Concede cleanly or not at all. A half-concession that hedges reads as evasion, which the ICLR data lists among the negatives. If a coverage claim is thin, say it is thin and say what will fix it.

1.6 Sources

- Noble, *Ten Simple Rules for Writing a Response to Reviewers*, PLOS Computational Biology, 2017.
- Maarten Sap, *Writing rebuttals*, personal notes, maartensap.com/notes.
- *Rebuttals Move Peer-Review Scores, but Initial-Review Structure Bounds the Movement*, arXiv 2606.22166, 2026.
- *Insights from the ICLR Peer Review and Rebuttal Process*, arXiv 2511.15462, 2025.
- *What makes a successful rebuttal in computer science conferences?*, Journal of Informetrics, 2023.

2 Machine learning venues

2.1 Where the practice differs

Machine learning venues run a longer, noisier, more interactive response process than security venues, and the advice that circulates in that community reflects it.

Length. NeurIPS, ICML, and ICLR allow far more than 750 words, and ICLR allows a threaded discussion rather than a single submission. Responses there routinely include new tables. A security response has to pick.

Interaction. At ICLR the exchange continues, and the empirical finding is stark: reviews with at least one author reply rose 27.0 percent of the time against 2.7 percent for reviews with none. Where a venue permits a second turn, taking it is most of the value. ACSAC does not, so the single submission has to carry everything.

New experiments. The expectation at machine learning venues is that a rebuttal can contain results run during the window. That expectation leaks into how reviewers there read a promise: if you could have run it and did not, the promise reads as an admission. Security reviewers, whose venues have explicit revision machinery, read promises more generously, but only when they are specific.

2.2 The request you cannot satisfy in the window

This is the case worth having a rehearsed answer for, because it arrives in almost every review.

What the evidence says not to do: promise vaguely, treat the request as unreasonable, or answer a different question. Vague promises about future work and superficial treatment of a concern are both validated negatives.

What works, in descending order of strength:

1. **Run it.** Baseline completion carries the largest positive odds ratio in the ICLR feature set, 1.64. If the experiment fits in the window at all, it beats every rhetorical alternative.
2. **Show that the question is already answered by something in the paper.** A missing baseline is sometimes a baseline the reviewer did not recognize. Naming the equivalence, for instance that an ablation configuration *is* the standard baseline being asked for, completes the baseline without running anything.
3. **Give a partial result or a bound.** An upper bound computed from numbers already in the paper is weaker than a measurement and much stronger than a promise, and it can be checked on the spot.
4. **Commit specifically.** Name the configuration, the table it will appear in, and the window it will appear by.
5. **Narrow the claim instead.** If the evidence supports something smaller than what the paper claims, saying so and rewriting the claim removes the objection permanently. This is unpopular and undervalued. It converts a dispute about evidence into a promise about wording, which is cheap to keep.

2.3 What the ceiling means for effort allocation

Initial score position dominates every feature of the exchange, at odds ratio 0.141 per standard deviation. A reviewer far from the panel consensus rarely moves regardless of what the response says.

The practical reading is not fatalistic. It says to spend the words where the distance to the line is small, and to accept that a determined outlier in either direction is mostly fixed. It also says that a paper with one weak reject and two positives is exactly the configuration where a response pays, since the weak reject has the most room and the panel has the most reason to look.

One caveat on the numbers. The 45.6 percent figure for substantively engaged exchanges is inflated by selection, since authors engage more when they think they can win. The honest figure to plan against is the 21 percent overall increase rate, against 1.4 percent decrease.

2.4 Sources

- *Rebuttals Move Peer-Review Scores, but Initial-Review Structure Bounds the Movement*, arXiv 2606.22166, 2026.
- *Insights from the ICLR Peer Review and Rebuttal Process*, arXiv 2511.15462, 2025.

- *What makes a successful rebuttal in computer science conferences? A perspective on social interaction*, Journal of Informetrics, 2023.
- *Testing for Reviewer Anchoring in Peer Review: A Randomized Controlled Trial*, arXiv 2307.05443, on why initial position dominates.
- Maarten Sap, *Writing rebuttals*.

3 Security venues

Try to find more tips specifically for ACSAC. Try to find more tips for the security venue in general. In addition to tips on what to write, we also need tips on how to interpret the reviews.

3.1 What the chairs ask for

The instruction is unusually consistent across the top security venues, and it is narrower than what journal advice suggests.

IEEE Symposium on Security and Privacy tells authors to focus *exclusively* on factual errors in the reviews and on concrete questions the reviewers asked, and adds that new results may be discussed when they clarify an open question. USENIX Security uses close to the same wording across several years: focus on questions posed by reviewers and on significant factual corrections. ACSAC 2026 phrases it as primarily addressing specific questions or concerns, and as clarifying misunderstandings or correcting misconceptions.

A general defense of the work is off-brief. The IEEE Symposium on Security and Privacy 2026 call is explicit: “The authors should mainly focus on factual errors in the reviews and concrete questions posed by the reviewers,” adding that “new research results can also be discussed if they help to clarify open questions.” The same call attaches a penalty to ignoring the instructions: “Failure to follow the instructions sent at the beginning of the rebuttal (for example submitting rebuttals over the word count limit) will result in immediate rejection.”

So restating the contribution, arguing that the problem matters, or explaining why the design is novel is not merely wasteful. It is outside what the venue asked for, and at one venue the instructions carry a stated consequence. The ICLR data agrees from the other direction: weak novelty justification is a validated negative.

Scope is limited; depth is not. This distinction is worth holding onto, because the instruction is easy to over-read. What the venues restrict is which comments a response addresses and how many words it spends. Nothing in any of them says the answers must be shallow. A factual correction that reaches the root of a problem is still a factual correction, and it is worth more than one that fixes the symptom. Where a reviewer’s objection is fundamental, an answer pitched at the surface reads as the failure mode the ICLR analysis calls inadequate explanation, which carries the largest negative odds ratio in the set at 0.31.

A factual correction is the highest-value move available. It is the one thing every security venue explicitly invites. A reviewer who has misread a number or a table is asking, without meaning to, for the cheapest possible score movement.

Questions are obligations. A question a reviewer wrote and the response did not answer is the clearest form of the failure mode the ICLR study calls superficial concern addressal.

3.2 Word limits and the shape they force

IEEE S&P and ACSAC both cap the non-interactive response at 750 words. That is roughly one and a half pages of prose, or eight short paragraphs. With three reviews and twenty or more comments, per-comment answers are not affordable.

The forced structure is thematic: group comments that share a root, answer the group once, and let the presentation items collapse into a single closing paragraph. Reviewers can still find their point if the grouping is named in their own vocabulary.

IEEE S&P also runs an interactive variant for some papers, where authors and reviewers exchange messages and not every reviewer participates. ACSAC 2026 does not; the response is one submission.

3.3 The revision path changes what a promise is worth

This is where security venues diverge most from machine learning venues, and it is the reason the ICLR finding on vague promises needs local adjustment.

ACSAC 2026 offers three outcomes: Accepted, Minor Revision, Rejected. Minor Revision is a conditional acceptance with an assigned program committee shepherd who guides the revision and leads the discussion of the revised paper. The window runs from notification on 8 September to 8 October, with camera ready on 22 October.

A promise made in the response therefore has somewhere to land. The committee is not being asked to take it on faith; they are being asked whether the work is shepherdable. That reframes the drafting question from *can we prove this now* to *is this specific enough that a shepherd could check it in four weeks*.

The corollary is that an unspecific promise is worse here than elsewhere, because the reader is actively evaluating whether the promised revision is bounded. “We will improve the evaluation” gives a shepherd nothing to hold. “We will run the fourth ablation cell and report the full two-by-two” is a task with an end.

USENIX Security runs a different model again, with major revision cycles across submission rounds, and IEEE S&P with monthly deadlines and revise-and-resubmit. In all three the same principle holds: find out what the venue’s revision machinery is, and write promises the machinery can accept.

3.4 What security reviewers read as evasion

Drawn from venue instructions and from the failure modes in the ICLR analysis, adapted to this community:

- Answering a methodological objection with a presentation fix. If a reviewer questions whether a comparison is fair, promising to rewrite the sentence that describes it does not answer the question.

- Pointing at a section number instead of stating the content. The reviewer has already read the section; that is how they formed the objection.
- Treating a threat-model gap as out of scope without saying what the scope is and why the boundary sits there.
- Claiming an artifact settles a question the response does not answer. If the artifact is anonymised and linked, a reviewer or an artifact evaluator can read it, so a claim about the implementation should match the code.
- Silence on a reviewer’s question, however minor.

3.5 Sources

- ACSAC 2026 call for papers and the program chairs’ notification of 19 August 2026.
- IEEE Symposium on Security and Privacy 2025, 2026, and 2027 calls for papers.
- USENIX Security submission policies and instructions, 2022 to 2024, and the messages from the program co-chairs for 2022 and 2024.
- *Rebuttals Move Peer-Review Scores*, arXiv 2606.22166, 2026, for the failure modes.

4 The structure of an argument

Booth, Colomb, Williams, Bizup, and Fitzgerald treat a research argument as a conversation rather than a proof. Part III of *The Craft of Research* (4th edition, University of Chicago Press, 2016) sets out five elements and the reader questions each one answers. The frame is useful twice over. It is a checklist for a draft, and it is a diagnostic for a review, because a reviewer’s objection is always an objection to one of the five.

4.1 The five elements

Chapter 7 opens with the summary sentence: “In a research argument, you make a claim, back it with reasons supported by evidence, acknowledge and respond to other views, and sometimes explain your principles of reasoning” (7.1, p. 110). The authors then derive the five elements from the questions a skeptical listener asks (7.1, p. 111):

1. **Claim.** What do you want me to believe? What’s your point?
2. **Reasons.** Why do you say that? Why should I agree?
3. **Evidence.** How do you know? Can you back it up?
4. **Acknowledgment and response.** But what about ...?
5. **Warrant.** How does that follow? What’s your logic? Can you explain your reasoning?

The definitions are given in a boxed note on p. 112. A claim is “an assertion (which can be a single sentence or more) that demands support.” A reason is “an assertion that supports a claim.” Evidence is “data deployed to support a reason,” and, unlike the other two, evidence is not always framed as an assertion. The core of the argument is then written as a single line: claim because of reason based on evidence (7.2.2, p. 114).

Two properties of a claim are assessed separately from its support. The claim must be specific, and it must be significant. Significance is defined operationally: it is “a quality they measure by how much it asks them to change what they think,” and the estimate is the number of other beliefs a reader would have to revise on accepting the claim (8.2.2, p. 127).

4.2 Reasons are not evidence

The distinction is the one writers most often collapse, and the book spends a full section on it because collapsing it produces arguments that look supported and are not.

The authors concede that the two words are interchangeable in some contexts. Their test is a pair of sentences (7.2.2, p. 113). “What evidence do you base your reason on?” is well formed. “What reason do you base your evidence on?” is not. Two further differences follow. “We use our minds to think up reasons.” Evidence has to be searched for “out there” in the world and then made available for everyone to see. Hence the diagnostic: “It makes no sense to ask, Where do I go to see your reasons? It does make sense to ask, Where do I go to see your evidence?”

The reason writers conflate them is that a chain of reasons feels like support. A reason is also a subclaim, and a subclaim can itself take reasons, so the chain can run for some distance without ever touching data. The authors state the stopping condition directly: these chains do not go on forever, and eventually you have to show some data.

Where the chain stops is not the writer’s decision. Chapter 9 makes the boundary a property of the audience: “To count as evidence, a statement must report something that readers agree not to question, at least for the purposes of the argument” (9.2, p. 133). If readers do question it, what the writer took for hard evidence is only a soft reason, and the argument has not reached bottom. A sidebar on the same page puts it as a rule: evidence counts as evidence only if readers accept it without question, at least for the moment.

Chapter 9 adds a complication that matters for empirical work. A paper almost never contains the evidence. It contains a report of the evidence: a table, a figure, a summary statistic. Data taken from a source have already been shaped by that source, and reporting them again shapes them once more, if only by placing them in a new context. The recommendation is to state how the data were collected and by whom, and to cite a source as close to the evidence as possible.

The criteria for evidence are a list of five (9.4, p. 138). Readers want evidence to be accurate, precise, sufficient, representative, and authoritative. Relevance is treated separately, because relevance is a question about the warrant.

4.3 Warrants

A warrant is “a principle that connects a reason to a claim” (chapter 11, p. 155). It exists because a reader can grant that a reason is true and still deny that it is relevant. The authors’ running example is a reader who accepts that the line of hard freezes is moving north, accepts that health

care costs are rising, and asks what the first has to do with the second (7.4, p. 115).

A warrant has two parts, a general circumstance and a general consequence, and its logic is that if a generalization holds, so must specific instances of it. For the inference to go through, a reader has to accept four things: that the warrant is reasonable, that the reason is true, that the specific circumstance in the reason is a plausible instance of the general circumstance, and that the specific consequence in the claim is a plausible instance of the general consequence (7.4, p. 116). Most academic warrants are stated in compressed form, with circumstance and consequence run together, but both parts can always be recovered.

When to state one. The governing rule appears in the chapter abstract and again in its first page: “state your warrants only if your readers will not be able to understand your reasoning without them, or if you anticipate that your reasoning will be challenged” (chapter 11, p. 155). Section 11.5 (p. 164) names three occasions. The readers are outside the field. The principle of reasoning is new or controversial within the field. The claim is one readers will resist because they do not want it to be true, in which case stating a warrant they already accept, before the reason and claim, is the recommended order.

The converse is equally explicit. Experienced researchers writing for specialists leave most warrants unstated, because stating the obvious reads as condescension rather than help (11.2, p. 158). This is why arguments written for specialists are hard for newcomers to follow: the connecting principle is present and invisible.

Five tests. Before a reader challenges a warrant, the writer should ask whether it is reasonable, whether it is sufficiently limited, whether it is superior to competing warrants, whether it is appropriate to this field, and whether it is able to cover the reason and claim (11.4, p. 161). The last test is the one that catches bad arguments that look valid. The authors work an example on violent video games in which the warrant is acceptable, the reason is true, and the argument still fails, because the stated reason is not an instance of the warrant’s general circumstance (11.6, p. 166).

Prefer evidence. The closing Quick Tip warns against building on warrants when data are available. A claim of fact cannot be supported by a warrant and a reason alone. The instruction is to rely “not on elaborate lines of reasoning based on warrants but on hard evidence” (Quick Tip, p. 172).

4.4 Acknowledgment and response

This is the element that separates an argument from an assertion. The chapter 10 abstract states it as a condition: “An argument is not complete if it fails to recognize other points of view” (p. 141). Chapter 7 says the same thing about the diagram: no research argument is complete without acknowledgments and responses (7.3, p. 115). A text that offers only claims, reasons, and evidence may still read as thin, or as ignorant of or dismissive of the views it did not name.

The authors sort anticipated objections into two kinds (p. 142). Objections to **intrinsic** soundness challenge the clarity of the claim, the relevance of the reasons, or the quality of the evidence. Objections to **extrinsic** soundness ask the writer to consider alternatives: a different framing of the problem, evidence that was overlooked, or what others have written. The second kind is usually found by reading sources for disagreement rather than for data, including disagreements among the sources themselves (10.2, p. 145).

Section 10.1 (pp. 143 to 144) lists the questions a hostile reader asks of the support. One asks for

a different kind of evidence. The rest attack quality: it is not accurate, not precise enough, not current, not representative, not authoritative. The authors identify the seventh as the hardest to answer, and it is the one that asks for more of it.

A response is not a competing claim. Even a minimal response supplies a reason for limiting or rejecting what was acknowledged, and a substantial one is a subordinate argument with its own reasons, evidence, and possibly its own warrant (10.4, p. 148).

Two further points are worth carrying into a rebuttal. First, acknowledging a flaw that cannot be fixed is recommended over concealing it, with three usable forms of response: that the rest of the argument outweighs the flaw, that the flaw is serious but further research will find a way around it, or that the flaw prevents full acceptance of the claim while the argument still shows what a better answer would need (10.3.2, p. 147). Second, on how to phrase disagreement: “Save criticism for the response, and direct it at the work rather than the person” (10.5.1, p. 150).

Selection is a real constraint. Section 10.3 (p. 145) calls it a Goldilocks problem. Too many acknowledgments distract from the core; too few make the writer seem indifferent to the reader’s views. The priority list is plausible charges the writer can rebut, alternative lines of argument important in the field, alternative conclusions readers want to be true, alternative evidence readers already know, and counterexamples that have to be addressed.

4.5 Testing an argument by asking a reader’s questions

The book’s method for testing a draft is to run the five questions against it before a reader does. Section 10.1 gives the procedure. Write the core first, then question it as someone with a stake in a different outcome would. The suggested posture is not warfare but a warm debate, adopted on behalf of a reader who wants the argument to be wrong.

The questions come in two batches. The first is about the problem and the solution: whether there is a problem at all, whether it has been properly defined, whether the solution is practical or conceptual and whether it matches the kind of problem, whether the claim is stated too strongly, and why this solution is better than the alternatives. The second is about the support, and is the list of evidence objections above. The instruction attached to the first batch is worth quoting for its cost logic: the time to fix a problem with an argument is when you find it.

The Quick Tip after chapter 10 (pp. 153 to 154) adds three disagreements that at least some readers always produce. There are causes in addition to the one claimed. There are counterexamples. And, third, “I don’t define X as you do.” On the last, the authors note that definitions function as warrants, so a dispute over a definition is a dispute over the connection between a reason and a claim, not a matter of vocabulary.

4.6 Reading a reviewer comment as an attack on one element

A review is the reader’s half of the conversation, delivered after the fact. Every substantive objection lands on one of five places. Identifying which one is faster than parsing the comment sentence by sentence, and it determines what a response can consist of.

Target	Form of the objection	What the response must supply
Claim	Overstated, vague, or not significant	A narrower or more precise claim, or an argument that acceptance would change what readers believe
Evidence	Not accurate, precise, sufficient, representative, or authoritative	Data, or an account of collection, or a stated limit
Warrant	The reason is granted but read as irrelevant	The connecting principle, stated, and a check that reason and claim are instances of it
Definition	A term is used in a sense the reader does not hold	The definition, with a subordinate argument for it
Significance	So what	The consequence of the result for what readers currently think or do

Three consequences follow.

An objection to the warrant cannot be answered with more evidence. The reviewer has already granted the reason. Adding a table addresses a question that was not asked. The response has to state the principle that makes the reason relevant, then show that the reason and the claim are good instances of it. If neither can be shown, the argument needs revision rather than defense (11.6, p. 168).

An objection to the evidence cannot be answered with more reasoning. This is the same rule read in the other direction. A claim of fact does not follow from a warrant and a reason. If a reviewer says the evaluation is thin, the answer is a run, a number, or a stated limit.

An objection to a definition is an objection to a warrant. Reviewers frequently write that a word in the paper does not mean what the paper takes it to mean. Treating that as a terminology complaint understates it. The definition licenses the inference, so the response owes a definition and a short argument for choosing it, including an acknowledgment of the common or competing sense and a reason for not using it.

One more use of the frame. When the same objection is raised by two reviewers in different words, sorting both by target shows whether it is one objection or two. Two comments that both target the evidence can be answered once. A comment on the evidence and a comment on the warrant cannot, however similar their wording.

4.7 Sources

- Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, Joseph Bizup, and William T. FitzGerald, *The Craft of Research*, 4th edition, University of Chicago Press, 2016. ISBN 978-0-226-23973-6. Every quotation and page reference above was read from the copy at `_side/_tips/the-craft-of-research-4th.pdf`. Page numbers are the printed page numbers in the running heads, not the page numbers of the file.
- Sections used: chapter 7, *Making Good Arguments: An Overview*, pp. 110 to 121; chapter 8, *Making Claims*, pp. 122 to 131; chapter 9, *Assembling Reasons and Evidence*, pp. 132 to

140; chapter 10, *Acknowledgments and Responses*, pp. 141 to 152, with the Quick Tip *Three Predictable Disagreements*, pp. 153 to 154; chapter 11, *Warrants*, pp. 155 to 170, with the Quick Tip *Reasons, Evidence, and Warrants*, pp. 171 to 172.

The table and the three consequences in the previous subsection are not in the book. They apply the book's five elements to reviewer comments, which is an extension the authors do not make.

5 Reading papers

Two sources are named on the Read shelf of the site, and they answer different questions. Keshav gives a procedure for deciding how much of a paper to read. Eisner gives a posture for reading it once that decision is made. A third method is already in use in this project, the twenty five question scheme behind `Literature/reading_insights.md`, and it does something neither of the other two does.

5.1 Keshav: three passes, and the decision to stop

The method exists to prevent one failure, stated in the abstract: reading is “rarely taught, leading to much wasted effort.” The structure is a sequence of passes rather than a single read, and each pass ends in a decision about whether to continue.

The first pass is “a quick scan to get a bird’s-eye view of the paper” and “should take about five to ten minutes.” Five steps: read the title, abstract, and introduction carefully; read the section and subsection headings and ignore everything else; glance at the mathematical content to determine the theoretical foundations; read the conclusions; glance over the references, ticking off the ones already read.

At the end of it the reader answers the five Cs, quoted in full:

1. **Category.** “What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?”
2. **Context.** “Which other papers is it related to? Which theoretical bases were used to analyze the problem?”
3. **Correctness.** “Do the assumptions appear to be valid?”
4. **Contributions.** “What are the paper’s main contributions?”
5. **Clarity.** “Is the paper well written?”

The pass ends in a stopping decision, and the paper names three legitimate reasons to stop: the paper does not interest you, you do not know enough about the area to understand it, or the authors make invalid assumptions. One pass is called “adequate for papers that aren’t in your research area, but may someday prove relevant.”

The second pass reads “with greater care, but ignore details such as proofs,” and “should take up to an hour for an experienced reader.” Two instructions carry it. Look carefully at the figures: “Are the axes properly labeled? Are results shown with error bars, so that conclusions are statistically

significant? Common mistakes like these will separate rushed, shoddy work from the truly excellent.” And mark relevant unread references for further reading. The paper quotes a suggestion from Dominik Grusemann to “note down terms you didn’t understand, or questions you may want to ask the author,” and observes that for a referee those notes become the review and the material to defend it with at the committee meeting.

The test of the second pass is stated as an ability rather than a feeling: “You should be able to summarize the main thrust of the paper, with supporting evidence, to someone else.”

When the second pass fails, the paper names the causes and the three options. Causes: unfamiliar subject matter and terminology, a proof or experimental technique the reader does not know, poor writing with unsubstantiated assertions and forward references, or the hour. Options: set it aside, return after reading background material, or persevere into the third pass.

The third pass has one governing idea. “The key to the third pass is to attempt to virtually re-implement the paper: that is, making the same assumptions as the authors, re-create the work. By comparing this re-creation with the actual paper, you can easily identify not only a paper’s innovations, but also its hidden failings and assumptions.”

The instruction that follows is the one that matters for review work: “You should identify and challenge every assumption in every statement.” The pass takes “many hours for beginners and more than an hour or two even for an experienced reader,” and ends when the reader can reconstruct the structure of the paper from memory.

5.2 Keshav: the literature survey, in three steps

The same paper gives a procedure for entering an unfamiliar area, and it is a concrete alternative to searching by keyword.

Step one. Find five recent highly cited papers in the area, do one pass on each, then read their related work sections. If that yields a recent survey, “Read the survey, congratulating yourself on your good luck.”

Step two. “Find shared citations and repeated author names in the bibliography. These are the key papers and researchers in that area.” Download those papers. Then look at where those researchers have published recently, which identifies the top venues in the field.

Step three. Go to the recent proceedings of those venues and scan for related work. Those papers plus the ones set aside are the first version of the survey. Make two passes through them, and if they all cite a paper not yet found, obtain it and iterate.

The procedure is worth noting for what it does not use. It does not start from a query. It starts from a small seed set and follows citation structure outward, which is the same operation as the bidirectional citation chasing in the literature review procedure.

5.3 Eisner: how to read once the decision is made

Eisner’s advice is about posture, and four instructions are usable directly.

Skim first, “skipping over anything that would take much mental effort,” to establish direction and novelty, then return and read carefully. The order is the same as Keshav’s, arrived at independently.

Write while reading. “Write as you read. This keeps your attention focused and makes you engage with the paper.” Two levels of note are recommended: low level annotations on specific content, and a high level summary for future retrieval.

Read creatively. Identify the good ideas, consider applications the authors did not, and ask what you would do next. Question what is unclear, fill in missing steps, and connect the paper to other work.

When stuck, change material rather than pushing harder. “When the going gets tough, switch to background reading,” meaning textbooks, tutorials, or review articles rather than more of the same dense original.

On selective reading, Eisner permits skimming a technical section when it explains familiar material, is overly specialized, or is too advanced, provided the reader still grasps what the method accomplishes.

5.4 The scheme already in use here

The twenty five question scheme behind `reading_insights.md` is a third kind of reading, and it should not be confused with the other two. Keshav and Eisner describe reading a paper. The scheme describes reading a paper for a purpose that is fixed in advance.

Its questions are not about the paper’s own argument. They are about what the paper supplies to a submission in progress: the Heilmeier framing, the single-sentence story, which contribution leads, who cares, reference template papers, section structure and page allocation, the twelve page section budget, threat model prominence, venue conventions, most cuttable content, failure modes, artifact sufficiency, where to publish, and so on. Every insight is recorded in the form *From <source>: ...*, one bullet per question per source, so that a later reader can collect one question across seventeen sources and see the spread of answers.

Three properties of that design are worth naming, because they are what make it work and they are reusable.

The question set is fixed before reading starts. The same twenty five questions are asked of every source. That is what makes the answers comparable across sources, and it is what allows a cross-source consistency pass to find contradictions.

Contradictions are recorded rather than resolved. The file’s own editorial section lists five framing contradictions across the seventeen sources and states that they “are EDITORIAL decisions the writer must make; they cannot be resolved by patching individual bullets.” A reading system that averages its sources loses this.

Quality is audited separately from insight. The `quality_*.md` files score each bullet for depth and check it against the source, which is a separate operation from generating the bullet. Self review by the same pass does not substitute for it.

5.5 What the measurement literature adds

Three findings change how much to trust a read, and all three concern the gap between reading an abstract and reading a paper.

Abstracts decide inclusion badly. Across 139,467 citations and 329,332 screening decisions by 86 reviewers, the total error rate for abstract screening was 10.76 percent. Of the citations screened,

18.07 percent passed the abstract stage and 5.48 percent survived to final inclusion. About two thirds of what an abstract admits is overturned once the body is read. A paper included on the strength of its abstract has not been read.

Extraction is accurate until the information is absent. A study of structured extraction reported 92.4 percent accuracy overall with a 95 percent confidence interval of 89.5 to 94.5, and false data produced in 5.2 percent of cases. The number that matters is the conditional one: reproducibility fell to 77.2 percent when the information was not reported in the paper, and the failures took the form of plausible but incorrect content. The lesson is that a question a paper does not answer is where fabrication enters, so *not stated* has to be an available answer.

An operating rule from a review of 172 studies. Mean extraction precision was 83.0 percent and recall 86.0 percent, and the policy the authors adopted is worth copying without change: for categories with precision below 80 percent, assign a manual reviewer to validate and correct. Performance metrics were among the lower accuracy categories, which is the category most likely to be reused.

5.6 Putting the three together

The three sources answer three different questions, and a reading session uses all of them in order.

Keshav decides how far to read, and gives a stopping rule at each depth. Eisner governs what happens during the read: write while reading, question actively, switch material when stuck. The fixed question scheme decides what to extract, and keeps the extraction comparable across sources so contradictions surface instead of averaging out.

The measurement literature adds one constraint that overrides all three. A claim sourced from an abstract is marked as such and is not treated as a claim about the paper. Where a question has no answer in the paper, the answer recorded is that the paper does not address it.

5.7 Sources

- S. Keshav, *How to Read a Paper*, ACM SIGCOMM Computer Communication Review 37(3), 2007. Quotations are from the version of 17 February 2016, read in full, text extracted locally from the file at `svr-sk818-web.cl.cam.ac.uk/keshav/papers/07/paper-reading.pdf` after a fetch summary failed to return the text. Sections 2.1 to 2.3 and 3.
- Jason Eisner, *How to Read a Technical Paper*, 2009, `cs.jhu.edu/~jason/advice/how-to-read-a-paper.html`. Read as a web page. Quotations are short and are marked where they are exact.
- The twenty five question scheme and its insight format are this project's own, in `Literature/reading_insights`. and the accompanying `PaperReads/quality_*.md` files.
- The abstract screening figures are from a study of error rates in abstract screening across 86 reviewers. The extraction figures are from a study of structured extraction and from a review of 172 studies. Both reach this document through the research agents and literature review note, which records the full citations and marks which were read in full.

The comparison in the final subsection is not in any of the sources. It sets three independent methods against each other, which none of the three authors does.

6 Talks and posters

The twenty minute conference talk is already specified in `internal/slides/TALK-RULES.md` on the site, distilled from Shewchuk and Peyton Jones, and that file governs. It is not repeated here. This section covers the formats the Presentation page lists and leaves empty: the short talk, the poster, and the reading group.

6.1 What the existing rules fix, so the rest can build on it

Four points from `TALK-RULES.md` carry into every format below, and they are quoted rather than restated.

The purpose: “An advertisement for the paper, not a replacement for it. The goal is that people leave wanting to read it.”

The four questions every deck answers: why the problem is worth solving; what the method accomplishes that no previous method does, *and* what idea makes that possible, with the note that “Most talks answer only the second”; what the evidence is and under what circumstances; and the one idea to take home, “One, not five.”

The shape: “Onion, not clew: state the main message early, then add layers, returning to it. A listener who drifts for a minute should still be able to rejoin.”

The rule on titles: “A title says what the slide argues, not what it contains: ‘Factoring keeps the adversary’s error contained’, not ‘Architecture’.”

6.2 The five minute talk

A five minute talk is not a twenty minute talk delivered faster. The constraint is that only two of the four questions survive at full length.

Inference, mine rather than the sources’. At five minutes, question one and question four are answered in full, question two is compressed to a single sentence naming the idea, and question three shrinks to one number under one condition. Setup, ablations, and limitations move to the backup slides and are reached only if asked.

That yields roughly five slides. The problem and why it matters. The idea, stated once, as a picture. The one result, one claim on the slide. The limit that a listener would otherwise raise as an objection. The one idea again, with where to find the paper.

Two consequences follow from the onion rule. The main message appears on the second slide at the latest, because there is no time for a listener to wait for it. And the last slide repeats the first, because at five minutes a listener has no other chance to recover the point.

The annotation rule from `TALK-RULES.md` matters more here, not less. Everything the speaker would say goes in the notes, in full sentences. At five minutes the temptation to put the script on the slide is strongest and the cost of doing it is highest.

6.3 The poster

A poster session is a filtering problem before it is a communication problem. A reader walking a hall decides in seconds whether to stop, and a poster that requires reading to make that decision has already failed.

The approach on record for fixing this is Mike Morrison's, published in 2019 as a template on the Open Science Framework with an accompanying video, and adopted widely enough to have its own literature of commentary.

Access note. The description below comes from secondary sources discussing the template rather than from Morrison's own template or video. It is marked as secondary and the primary is named, and a poster built from it should open the template first.

The structure inverts the usual one. The center of the poster carries a single plain-language statement of the main finding, in large type, on a solid background. The traditional sections remain present, in a narrow column at reduced size, so a reader who has stopped can find the method, the results, and the discussion. A code linking to the full work sits below the finding, so the detailed material is one scan away rather than on the board.

The design goal, as the commentary states it, is to maximize the insight a reader takes from a session rather than the information a single poster contains. The mechanism is self-selection: a reader decides in seconds whether the work is relevant, and the ones who stop are the ones worth talking to.

Two things carry over from the talk rules without change. The main finding on the poster is a claim, in the project's own terms, written the way a slide title is written. And nothing is asserted that the folder does not support, with a placeholder marked in the open where a result is not in yet.

6.4 The reading group

The Presentation page names one source for this and it is the right one. Cathy Wu's paper draws on fourteen reading groups across five departments, and it identifies a single technique as the thing that makes the format work.

The technique, quoted in full. "At the start of the meeting, ask/make/force everyone ask 2 questions at the beginning and write them on a white board if possible. At this time, welcome any and all questions. Try to prevent members of the group from jumping in and answering them right away because these questions form an outline for the discussion. Then, use the awkward silence to your benefit, to get more questions asked and put on the board."

Why it works is stated directly, and the reasoning is worth keeping: "With these questions, all the participants now have a stake in the discussion and wants their questions answered. Importantly, this approach communicates to participants that they are allowed to not understand parts of the paper, but that they must read carefully enough to have questions. This is the secret to making reading groups work."

The participation baseline. The paper gives a number worth knowing before judging a session: "In any reading group, you can expect a maximum of 50% to participate actively, 20% to talk sometimes, and 30% to never talk," attributed to ten or more years of reading group experience. Against that baseline the author reports 80 to 90 percent participation using the technique above.

A quiet room is the default, not a failure of the material.

Facilitation rules, quoted. Encourage everyone to talk, with the target that participation rises across a term rather than falls. “Ensure that no particular group member dominates the discussion. This includes you, of course.” Maintain lexical order, going from the beginning of the reading to the end, avoiding arbitrary skipping, while keeping the last part of the discussion free for details or broader subjects. “Awkward silence is an indication to go onto the next question on the board.” Keep a machine available to look up background the group does not have. Invite a professor or research scientist where a discussion would benefit from it.

Opening and closing. The first question “can simply ask for the big picture and summary of the reading material.” The closing one: “A good closing question is what the group thought of the paper. This is also a great way to get feedback on your material selection, topic choice, discussion, etc.”

The facilitator’s own position. “As a facilitator, you should be more familiar with the papers than if you were simply in the group. That said, you don’t need to know everything. Just like everyone else in your group, you should not hesitate to ask any questions you may have.”

Two failures the paper records from experience. A group without a stated goal drifts, and the fix is to say at the start of a meeting what the group should be trying to do and why, with the note that participants “need not reach the goal while reading the paper, but the goal will give the entire discussion a general focus.” And a group where discussion comes from a few students, where the recommendation is to keep pressing everyone to speak while watching for the person speaking too much.

Selecting the reading. Aim for eight to forty pages, on the reasoning that shorter readings do not sustain an hour and a half and longer ones are not read. Choose material accessible to the prerequisites set for the group, with the caution that “It is often easy to forget how much you may know about a subject.” The paper also reproduces Keshav’s three step survey procedure for finding material, which is the same procedure recorded in the reading tip.

6.5 Formats still without a source

The Presentation page lists two more headings with nothing under them, and neither is written here because no source has been read for them.

The job talk. The seminar and the lecture.

Both are on the artifact list rather than filled in with generic advice. The rule from `TALK-RULES.md` applies to this file as much as to a deck: nothing is asserted that the folder does not support.

6.6 Sources

- `internal/slides/TALK-RULES.md`, in the site repository. Read in full. Its own sources are Jonathan Shewchuk, *Giving an Academic Talk*, and Simon Peyton Jones, *How to Give a Great Research Talk*, neither of which was opened for this file.
- Cathy Wu, *How to Lead a Technical Reading Group*, MIT Computer Science and Artificial Intelligence Laboratory. Read in full, text extracted locally from the file at `wucathy.com/assets/pdfs/wu2012how.pdf` after a fetch summary failed to return it. Sections IV to VIII. The participation figures in section

VI are attributed there to Professor Polina Golland, and the white board technique to the same source.

- Mike Morrison, the Better Poster template, 2019, distributed through the Open Science Framework with an accompanying video. **Not retrieved.** The description here is from secondary commentary and should be checked against the template before a poster is built from it.

The five minute structure in the second subsection is an inference from the four questions in TALK-RULES.md. No source states it.

7 Answering three reviewers at once

7.1 Who actually reads the response

The response is addressed to three reviewers, but it is read in a discussion none of the authors can enter. Area chairs, or program committee members in that role, encourage reviewers to discuss a submission with each other and with the chair, in notes visible only to them, and they attempt to reconcile disagreements between reviews.

Three consequences follow, and they are the reason a multi-reviewer response is not three single-reviewer responses stapled together.

Everything is read by everyone. A concession made to one reviewer is read by the other two. So is a refusal. A response that tells Reviewer A the environment is more complex than it looks, and separately tells Reviewer C the evaluation is thin, has to survive both being read side by side.

The chair is synthesizing, not adjudicating line by line. What reaches the decision is a summary of whether the objections were met. Structure that a chair can skim beats prose that rewards close reading.

The authors are absent from the conversation that decides. Whatever the response does not settle gets settled without them. That argues for closing questions rather than opening them, and for not leaving a disagreement between reviewers for the panel to resolve unaided.

7.2 Shared concerns are worth naming as shared

When a criticism is raised by more than one reviewer, saying so in the response is standard practice and does real work. It signals that the reviews were read together rather than triaged separately, and it lets one answer discharge two obligations without appearing to skip one.

It also changes the arithmetic of the word budget. Four comments in this set say the evaluation is narrow. Answering them once, and saying that is what is happening, costs a fraction of answering them four times.

7.3 Disagreements between reviewers

The harder case, and the one with the most upside.

When two reviewers read the same thing differently, the response can use the disagreement rather than tiptoe around it. Systematically cataloging the points from every review is what surfaces these; indirect contradictions between reviewers are easy to miss when reviews are read one at a time.

Two moves are available.

Cite the other reviewer. Where Reviewer A infers that the environment must be simplistic and Reviewer C credits the same testbed as substantially more realistic than a symbolic simulator, C's assessment is evidence, and it costs nine words to invoke. It is also more persuasive than the authors asserting the same thing, because it comes from inside the panel.

Do not adjudicate between them. Saying one reviewer is right and another wrong asks the panel to take sides in a dispute the authors started. Supply the fact that resolves it and let them draw the conclusion.

7.4 Protecting goodwill

A response can lose ground it already held. Reviewer A calls the experimental evaluation strong, Reviewer B opens with very interesting and timely, and Reviewer C lists three explicit strengths. None of that has to be defended, and all of it can be undermined by a concession that goes further than the objection required.

The specific trap in this set: conceding to Reviewer A that the environment is simplistic would discard Reviewer C's endorsement of the testbed, which is the strongest positive statement in the reviews. The concession that A's comment actually requires is much narrower, about how the sample budget is reported.

7.5 Ordering

Nothing in the venue guidance requires answering reviewers in order, and there are better principles.

Weight by what can move. Initial score position bounds how far a review can shift, so words spent on a reviewer who is far from the line buy less than words spent on one near it.

Lead with what is answerable now. A correction or a completed baseline lands harder than a promise, and it sets the tone for how the rest is read.

Group by concern, not by reviewer. A paragraph per theme, naming which reviewers raised it, reads as a synthesis. A paragraph per reviewer reads as three separate conversations that happen to share a page.

7.6 Sources

- Noble, *Ten Simple Rules for Writing a Response to Reviewers*, PLOS Computational Biology 2017, on summarizing shared criticisms.
- Maarten Sap, *Writing rebuttals*, on citing other reviewers and on heading style.
- Dershowitz, *Rebutting Rebuttals*, Communications of the ACM, on how responses are actually consumed.

- ICML Area Chair Instructions, for how discussion among reviewers is run.
- Shah et al., *Design and Analysis of the NIPS 2016 Review Process*, arXiv 1708.09794.
- Matt Might, *Responding to peer review*.

8 How reviews are discussed and decided

Question 10 in the open questions asks for this directly: “Then try to play out potential discourses across the three reviewers, and the chair as well. This aspect is more subtle. Some research on this topic, and even specific research about paper review, rebuttal, and acceptance or rejection deliberation. I believe there is such research.”

There is. Three studies carry it, and two of them were run by program chairs on their own conference.

8.1 The consistency experiments

The design is the same in both years. A fraction of submissions is reviewed twice by two independent committees that do not know about the duplication, and the two outcomes are compared.

2021. “During the assignment phase of the review process, we chose 10% of papers uniformly at random,” assigning each two area chairs and twice the usual number of reviewers, with the committees split across two copies and unaware of it. 882 papers were duplicated out of 8,765 still under review.

The outcome: 203 of 882 papers, 23.0 percent, had inconsistent outcomes between the two committees. Excluding borderline cases the chairs marked as bumpable, the figure fell to 16 percent. And the number that matters for an author: “50.6% of papers accepted by the first committee were rejected by the second.”

When the chairs asked what the disagreements were about, the largest single answer was that there was nothing to disagree about. “48% of multiple-choice responses were ‘This was a borderline paper that could have gone either way; there was no real disagreement between the committees.’” One area chair called it noise on the decision frontier.

2014, revisited seven years later. Cortes and Lawrence went back to the first experiment with the citation record in hand. “We determine that 50% of the variation in reviewer quality scores was subjective in origin.” For accepted papers there was no correlation between quality scores and later impact measured by citations. For rejected papers, traced to wherever they were eventually published, there was a correlation.

Their conclusion is the sentence to keep: “the reviewing process for the 2014 conference was good for identifying poor papers, but poor for identifying good papers.”

They also warn against the obvious fix: they “warn against removing the subjective element.”

8.2 What moves during the discussion

The consistency experiments measure whether the outcome is stable. They do not measure what changes it. A process-centric study does.

Jung, Pyeon, Heo, and Ahn examined ICLR from 2017 to 2025, “encompassing over 28,000 submissions,” using the record that open review platforms create: “not only scores and comments but also rebuttals, reviewer-author exchanges, reviewer disagreements, and meta-reviewer decisions.”

Their finding on the response phase: “the rebuttal stage emerges as a decisive phase: timely, substantive, and interactive author-reviewer communication strongly increases the likelihood of acceptance, often outweighing initial reviewer skepticism.”

Alongside it, and in the same sentence class, they name “effective resolution of reviewer disagreement” as correlating with higher acceptance, together with clearer writing, balanced visual presentation, and earlier submission.

The dimension they measure for the committee side is “the patterns of reviewer disagreement and meta-review mediation (e.g., score variance, confidence weighting).”

8.3 The score movement figures already collected here

The tips work for this package recorded the movement data separately, and it fits the picture above. Most scores do not move: between 75 and 81 percent unchanged, 17 to 23 percent up, and about 1 percent down. But acceptance conditions sharply on movement: between 55.7 and 57.6 percent acceptance where a score moves, against 7.8 to 12.4 percent where none does.

Read with the consistency figures, that says the response is not competing against a fixed judgment. It is competing against a decision that was close to arbitrary in about half of frontier cases, where a single moved score changes the acceptance probability by a factor of five or more.

8.4 What follows for a response

Five consequences, and they are about who the reader is rather than about what to argue.

The reader who decides is the chair, not the reviewer. The chair mediates disagreement and weights by confidence. A response that answers three reviewers separately and never gives the chair a way to reconcile them has left the mediation undone.

A split panel is the leverage point, not the problem. Where two reviewers disagree about the same object, one of them has already made the argument. The wording checklist for this package states the move as a rule: use another reviewer’s assessment as evidence where they disagree with each other. It costs no words of the author’s own credibility.

At the frontier, the goal is to give a reason, not to win. Forty eight percent of disagreements were reported as having no real disagreement behind them. In that regime the response supplies the sentence a chair can quote when writing the decision.

Expertise is weighted, so it matters which reviewer holds which position. Confidence weighting is one of the measured dimensions. A concern raised by a self-declared expert and a concern raised by a reviewer with some familiarity do not carry the same weight, and a response that spends equally on both has misallocated.

The process detects bad work better than it detects good work. Cortes and Lawrence found the 2014 process good at identifying poor papers and poor at identifying good ones. The operational reading: removing a reason to reject is worth more than adding a reason to accept, because the process is more reliable in that direction.

8.5 The configuration in front of us

This package’s own panel is the frontier case the experiments describe. One accept from a self-declared expert, one weak accept from a knowledgeable reviewer, one weak reject from a reviewer with some familiarity.

Two of the comments are about the same object and point in opposite directions. One reviewer infers a simplistic environment from the small training budget. A different reviewer credits the same testbed for running VMs, real services, real network isolation, and correlated user activity, and for being explicit about the boundary between real and emulated components.

That pair is the disagreement the chair has to mediate, and the second reviewer has already written the answer to the first.

8.6 Sources

- The NeurIPS 2021 Consistency Experiment, posted on the conference blog by the program chairs, 8 December 2021. Read as a web page. All 2021 figures above are quoted from it.
- Corinna Cortes and Neil D. Lawrence, *Inconsistency in Conference Peer Review: Revisiting the 2014 NeurIPS Experiment*, arXiv 2109.09774. **Abstract only.** The two findings quoted are from the abstract. The disagreement rate for 2014 that circulates in secondary summaries is not in the abstract and is not repeated here.
- Sangkeun Jung, Goun Pyeon, Inbum Heo, and Hyungjin Ahn, *What Drives Paper Acceptance? A Process-Centric Analysis of Modern Peer Review*, arXiv 2509.25701. **Abstract only.** Every quotation is from the abstract, and the effect sizes behind the claims have not been checked.
- The score movement figures reach this file through the tips work already done for this package, which records the full citations.

The five consequences and the reading of this package’s own panel are extensions. None of the sources discusses this submission.

9 Problem finding

The Think shelf names three sources and all three are about solving rather than finding. Pólya gives a procedure for working a problem once you have one. Poincaré describes the two faculties a mathematician works with. Tao describes when intuition becomes trustworthy. None of the three tells you which problem to take up.

That gap is the subject here, and the instrument for it already exists in this project. The Headspace page is a problem-finding artifact, and its structure is worth naming so it can be used deliberately.

9.1 Why a list of heuristics is not the deliverable

The Mental World Modeling thread states the obstacle in its own words, and it applies to problem finding as much as to problem solving.

“General heuristics (‘work backwards,’ ‘try a simpler case’) transfer badly: people who can recite them still fail to deploy them, because knowing an operation and regulating when to use it are different capacities.”

The Next page names the same thing as the under-modeled part: “The evidence is consistent that people fail less for lack of heuristics than for lack of regulation, choosing the next move, noticing a line isn’t working, deciding when to abandon it. That control layer is a distinct problem from the operations it governs, and it is the under-modeled one.”

So this section gives a procedure with stopping conditions rather than a list of moves. The stopping conditions are the part that carries the work.

9.2 Poincaré: which faculty is being used

Poincaré separates two kinds of mind and assigns each a job. The distinction is useful here because problem finding and problem solving draw on different ones.

“Thus logic and intuition have each their necessary role. Each is indispensable. Logic ... is the instrument of demonstration; intuition is the instrument of invention.”

The two work differently. Logical minds advance “step by step, after the manner of a Vauban who pushes on his trenches.” Intuitive minds make “quick but sometimes precarious conquests, like bold cavalrymen of the advance guard.”

Each has a failure. Analysis “can have, therefore, no tool other than the scalpel and the microscope,” yielding “a multitude of procedures” without showing “which will lead us most promptly to our goal.” Intuition alone “can not give us rigour, nor even certainty.”

The passage that matters most for reading and for reviewing: verification is not understanding. “When we have examined these operations one after the other and ascertained that each is correct, are we to think we have grasped the real meaning of the demonstration? Shall we have understood it ...? Evidently not.”

Problem finding is invention, so it runs on the faculty Poincaré says cannot give certainty. That is not a reason to distrust it. It is a reason to write the candidate down and test it rather than acting on it directly.

9.3 Tao: when your intuition is worth acting on

Tao’s three stages answer the question Poincaré leaves open, which is when an intuition has earned the right to direct work.

The pre-rigorous stage is one “in which mathematics is taught in an informal, intuitive manner, based on examples, fuzzy notions, and hand-waving,” with emphasis on computation rather than theory.

The rigorous stage is one “in which one is taught that in order to do maths ‘properly’, one needs to work and think in a much more precise and formal manner,” where “one is expected to be able to comfortably manipulate abstract mathematical objects without focusing too much on what such objects actually ‘mean’.”

The post-rigorous stage is one “in which one has grown comfortable with all the rigorous foundations

of one’s chosen field, and is now ready to revisit and refine one’s pre-rigorous intuition on the subject, but this time with the intuition solidly buttressed by rigorous theory.”

The sentence that decides what rigor is for: “The point of rigour is not to destroy all intuition; instead, it should be used to destroy bad intuition while clarifying and elevating good intuition.”

The Teach page already states the failure this framework names, and states it as a design commitment: “Stalling at the second stage, with intuition discarded as unrigorous, is the failure I design against.”

The consequence for problem finding. An intuition about which problem matters is worth acting on in proportion to how much formalism has already passed through it. An intuition that has survived formalization is one that can be built on. An intuition that has never met the formalism is a candidate, not a direction.

9.4 Pólya, and where his procedure starts

Pólya’s four phases are already in use in this project as a reading lens, applied across seventeen sources in `reading_insights.md`: understand the problem, devise a plan, carry it out, look back.

Two things about that use are worth extracting.

The phases are applied to *other people’s* papers, to see how each one decomposed its problem. That turns Pólya from a solving procedure into a reading procedure, and the reading is where the candidate problems come from.

The heuristic that does the most work in those notes is the one about recognition. Applied to prior cyber environments, it surfaces as a critique: previous work is faulted for unrealistic goals, engineered rewards, and leaking the action space. The note records the move as restating by contrast, and it is the operation that converts reading into a problem.

Pólya’s procedure begins after a problem is in hand. The look back phase is the one that produces new problems, and in this project it is instantiated as turning each design choice into a direct answer to a named prior failing.

9.5 The Headspace page, read as an instrument

The Headspace page has a fixed shape and the shape is the method. Each entry is a question in the author’s own words, with sources listed under it. Some questions carry sub-questions that narrow the parent.

Four properties make it work.

The question is written before the answer exists. Entries such as when is a cybersecurity decision a reinforcement learning problem, or what is mind and what is muscle, have no answer under them. They have sources that are relevant to them.

Sources are attached to questions rather than to topics. A paper appears under the question it speaks to. The same paper could sit under two questions, and where it does, that is a finding about the questions.

Questions are stated at a level where they could be wrong. When does an agent have the need to model the world in the first place is answerable. What is interesting about world models is not.

Sub-questions narrow rather than elaborate. Under the opponent modeling entry, the sub-questions ask what defenders want to predict about adversaries, what the reality of data availability is, whether zero-day defense is a zero-shot or a few-shot challenge, and whether existing opponent modeling can differentiate human from AI adversaries. Each one could be taken up as a project.

9.6 A procedure for locating a topic

Question 32 in the open questions asks to triangulate the pinpoints of cyber attacker modeling against opponent modeling advances and challenges, and the author's own note on it asks for a pseudocode of that operation. This is that procedure.

Inference. This is assembled from the sources above and from the Headspace structure. No source states it.

1. **Write the question so it could be wrong.** If no observation would settle it, it is a topic and not a question. Rewrite until it names something that could turn out either way.
2. **Name the two fields it sits between.** A question worth taking up usually sits where two literatures make different assumptions about the same object. Attacker modeling and opponent modeling are one such pair.
3. **Find what each field takes for granted that the other does not.** This is the census operation from the lexicon method, run across two fields rather than across one project. Where the two use one term in two senses, that is a candidate.
4. **Check the demand side.** Ask what a practitioner in the applied field wants predicted, and whether the data to predict it exists. The Headspace sub-questions do this explicitly: what do defenders want to predict about adversaries, and what is the reality of data availability.
5. **Check that the answer would change something.** From the wording checklist: a number earns its place by changing a decision. A question earns its place the same way. Name what would be done differently under each answer.
6. **Check whether it is reachable.** A question is reachable when an experiment you could run this year would move it. Otherwise it is a five year question and belongs on the vista rather than in an agenda.
7. **Stop when a round of reading adds no new candidate.** Same stopping rule as the existence search. If a complete pass over the two literatures produces nothing new, the space has been mapped, and what remains is choosing rather than searching.

The four tests in steps one, five, six, and seven are the regulation layer. They are what the Mental World Modeling thread says people lack, and they are the reason this is written as tests rather than as prompts.

9.7 Sources

- Henri Poincaré, *Intuition and Logic in Mathematics*, read at mathshistory.st-andrews.ac.uk/Extras/Poincar
Quotations are exact as returned by that page.

- Terence Tao, *There's more to mathematics than rigour and proofs*. **Primary not retrieved.** The site returned 403 on two attempts and the archive was unreachable. The three stage quotations here come from a search summary of that page and are marked secondary. They should be checked against the original before being quoted in anything public. The author's own restatement of the same framework is on the Teach page and is quoted from there directly.
- George Pólya, *How to Solve It*. Not opened for this file. The four phases and the recognition heuristic reach this document through their use as a reading lens in `Literature/reading_insights.md`, question 2, applied across seventeen sources.
- The Mental World Modeling overview and its Next page, and the Headspace page, all read in full from the site.

The procedure in the previous subsection is an extension. It applies the Headspace structure and the lexicon census to the problem of choosing a topic, which none of the named sources does.

10 Making reinforcement learning work, and reporting it

The site page for this is a stub, but the instrument already exists in this project. `Literature/PaperReads/matrix_en` reads twenty one sources against twenty six fixed columns, and those columns are the checklist. This section states what the columns are for, what the methodology papers found, and where they disagree.

10.1 The twenty six columns

The lens is applied to every source in the same order, so answers are comparable across papers. Grouped by what they check:

Statistical practice. Seeds. Variance reporting. Random source control. Statistical tests. Diverged-run handling.

Hyperparameters and convergence. Hyperparameter reporting. Hyperparameter sensitivity. Convergence demonstration.

Comparison. Calibration baselines. Baseline tuning. Implementation matching. Environment selection. Comparison style.

The problem as posed. Episode length. Discount factor. Reward function. Observation construction. Action space.

Learning machinery. Exploration. Credit assignment. Replay buffer. Offline learning support.

Evidence. Generalization. Ablation discipline. Sample complexity against compute. Action latency budget.

Two properties make it work. The columns are fixed before reading, so a paper cannot set its own bar. And *not reported* is a legal answer, so silence is recorded rather than filled in. Several entries in the matrix read “Not reported” or “Not discussed,” which is a finding about the field.

10.2 Seeds, and how many is enough

Four sources answer this and they do not agree on a number. All four agree that five is too few.

Henderson and colleagues, AAAI 2018, ran five trials per evaluation with preset seeds shared across conditions. Their demonstration is the useful part: they split ten trials into two groups of five, ran the same algorithm with the same hyperparameters in both, and found the two groups statistically different at $t = -9.0916$, $p = 0.0016$. Two identical configurations produced a significant difference through seed choice alone.

They also argue against a common practice: “it is not uncommon for the top- N trials to be selected from among several trials ... Our experiment with random seeds shows that this can be potentially misleading.”

Colas and colleagues treat the question directly. With $N = 5$ the bootstrap test produced a false positive rate near 10 percent against a nominal 5 percent. Welch’s t-test came in slightly above 5 percent. Their recommendations are concrete: use a larger sample than the power analysis prescribes, prefer Welch’s t-test to the bootstrap test, apply a Bonferroni correction for multiple comparisons, and set the significance level below 0.05 so the empirical type one error stays under 0.05.

Agarwal and colleagues, NeurIPS 2021 Outstanding Paper, ran 100 runs per algorithm and game across five algorithms and twenty six games, then subsampled at 3, 5, 10, 25, 50, and 100 to study how the conclusion moves. Their assessment of what the field would need: the number is “closer to 50–100 runs in Atari 100k – far too many to be computationally feasible for most research projects.”

They also identify a bias that matters at small sample sizes: “the sample median is a biased estimator of the true median ... in the few-run regime, we find that this bias can dominate the comparison between algorithms.”

Engstrom and colleagues, ICLR 2020, trained “at least 80 agents for each estimate (more for some high-variance cases),” with 95 percent confidence intervals from a thousand-sample bootstrap, and added agents where variance was high rather than reporting a wider interval.

The systematization of pitfalls in reinforcement learning for cybersecurity sets the bar at twenty runs with 95 percent confidence intervals, and records that a large share of surveyed papers fail at least one reporting criterion.

10.3 Where the sources disagree

One disagreement is unresolved and should stay that way rather than being averaged.

Colas recommends a significance test, names Welch’s t-test as the preferred one, and gives the correction and the significance level to use with it.

Agarwal recommends against significance tests: “we recommend using confidence intervals ... but avoid statistical significance tests (e.g., $p\text{-value} < 0.05$) because of their dichotomous nature ... and common misinterpretations.” In their place: stratified bootstrap confidence intervals, performance profiles, the interquartile mean as a robust alternative to the median, the optimality gap, and the average probability of improvement.

The empirical design source sides with intervals, recommending paired difference confidence intervals over hypothesis tests.

Both positions are defensible and they lead to different tables. Choose one, state which, and do not mix them.

10.4 What this project decided

The position on record in `reading_insights.md`, arrived at after the disagreement above was surfaced: five seeds, plus paired bootstrap confidence intervals, plus per-seed worst-case disclosure.

That is below the twenty run bar and far below Agarwal’s fifty to one hundred. The disclosure is what makes it honest. Raw per-seed episode losses ship with the artifact bundle so an independent bootstrap can be computed, which is stated in the paper’s evaluation setup.

Two facts about the current paper follow from the columns and are worth having in view. Five seeds is stated. Statistical tests are not reported, and the ordering claim rests on the gap exceeding one standard deviation in every cell.

10.5 Baselines, and the three ways a comparison fails

The columns separate three failures that are usually collapsed.

Calibration baselines ask whether a trivial policy would have done as well. A random policy and a do-nothing policy bracket the range and cost almost nothing to run. The current paper has three such references, Sleep, Random, and an expert-inspired heuristic, and the heuristic is the one that carries the argument because it encodes the standard playbook as a fixed policy.

Baseline tuning asks whether the comparison method was given the same care as the proposed one. The empirical design source names untuned baselines and designer bias as endemic. Henderson demonstrates the size of the effect by varying one hyperparameter at a time across four algorithms.

Implementation matching asks whether the two methods came from the same code. Henderson compared three codebases for one algorithm and three for another, and found the codebase itself changed the result. Engstrom took the stronger route, running everything from one codebase and listing every parameter, and states that it acknowledges its own bug.

The distinction that Engstrom isolates is worth carrying: code-level optimizations can account for a difference attributed to the algorithm. Their design separates the step method from the code-level options and runs all combinations, which is the only way to tell the two apart.

10.6 The nine challenges, and which ones a project addresses

Dulac-Arnold and colleagues name nine challenges of real-world reinforcement learning, each restated as a perturbation of the underlying decision process, each isolated, then recombined. The list is a yardstick a reader already knows.

The operative move for a paper is calibration rather than coverage. Name which challenges the work attacks, which it inherits, and which it puts out of scope. For this project the notes record the match as learning on a live system from limited samples, partial observability with a moving adversary, and the multiple objective tension between containment and keeping users working. Out of scope: low latency constraints, explainability, learning from logged data, and hard safety constraints.

Stating the out-of-scope list is the part that gets skipped and the part that prevents a reviewer from reading a partial result as a universal claim.

10.7 Two rules that apply before any of the above

Do not form the hypothesis after seeing the results. The empirical design source names this directly, and its recommended workflow puts the scientific question, the hyperparameter sets, and the seed budget in writing before the runs start. The look back step is honest re-examination rather than a narrative fitted to what came out.

Convergence goes in the body. A learning curve in an appendix is a claim the reader is asked to take on trust. Henderson also warns that a curve can hide a local optimum, so the curve alone is not sufficient evidence that the policy is doing what its score suggests.

10.8 Sources

- `Literature/PaperReads/matrix_empirical_rl_lens.md`, this project’s own instrument. Twenty six columns applied to twenty one sources. All quotations below reach this file through that matrix and its accompanying quality reports rather than through a fresh read of each paper, which is recorded here so the access level is not overstated.
- P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, D. Meger, *Deep Reinforcement Learning That Matters*, AAAI 2018.
- C. Colas, O. Sigaud, P.-Y. Oudeyer, *How Many Random Seeds? Statistical Power Analysis in Deep Reinforcement Learning Experiments*, arXiv 2018, with a later journal version.
- R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare, *Deep Reinforcement Learning at the Edge of the Statistical Precipice*, NeurIPS 2021, Outstanding Paper. Companion library `rliable` at github.com/google-research/rliable.
- L. Engstrom, A. Ilyas, S. Santurkar, D. Tsipras, F. Janoos, L. Rudolph, A. Madry, *Implementation Matters in Deep Policy Gradients*, ICLR 2020.
- G. Dulac-Arnold and colleagues, *Challenges of Real-World Reinforcement Learning*, Machine Learning Journal 2021, arXiv 2003.11881.
- The systematization of pitfalls in deep reinforcement learning for cybersecurity, and the empirical design source, both read for this project and recorded in **PaperReads**.

The grouping of the twenty six columns into six kinds is not in the matrix. The matrix lists them in order without grouping.

11 The other ACSAC formats

Everything below is for ACSAC 2026, held 7 to 11 December 2026 in Los Angeles. Venue policies are scoped to a year and this file is not valid for another one. Sources and access dates are at the end.

The refereed paper track and the 750 word response are covered elsewhere in these notes. This section covers the three other routes into the conference: posters and works in progress, panels, and case studies. Each has a different deadline, a different audience, and a different thing it publishes.

11.1 Posters and works in progress

Dates. “Deadline: September 19 (23:59 Anywhere on Earth).” Notification is “rolling.”

Length and format. “Poster proposals should be at most 2 pages, including references.” Submissions use “the 2-column IEEE template available at ieee.org/conferences/publishing/templates.html.”

Anonymity. The opposite of the paper track. “Poster proposals are not anonymous, so please make sure you include author information (name, affiliation, country) in your proposal.”

What it is for. The session “provides an opportunity for researchers and practitioners to present their new and innovative preliminary results and latest works-in-progress in an informal, interactive setting.” The stated benefit to the presenter is feedback at an early stage rather than a publication.

Cost and obligation. “If your proposal is accepted, a presenter is required to register, pay, and attend the conference.” The presenter must set up beforehand and stand with the poster for about 75 minutes. One concession: “accepted Poster authors do not need to pay the ‘Publication Surcharge’ that is requested of authors of accepted technical papers.”

What the two page limit implies. Two pages including references is not a short paper. It holds the objective, what has been done, and what comes next, and nothing else. The Better Poster structure in the talks section applies to the board itself rather than to this proposal.

11.2 Panels

The panel track runs on a much earlier schedule than the others, and the schedule is the thing most likely to be missed.

Dates. “Abstract Deadline: July 8.” “Notification: July 15.” “Complete Panel Details: August 20.” A final description, the panelists and their affiliations, and one paragraph statements from each panelist are due by 1 September.

Length. The proposal is “no longer than one page,” carrying a title, the moderator, the potential panelists, and “a brief one or two paragraph description.”

What the chair wants. Panels should “encourage audience participation and focus on the sharp edges of a new and emerging topic where there is controversy or where there are widely varying positions.”

That sentence is the selection criterion. A panel proposed around a topic where the panelists agree is proposing a tutorial. The proposal has to name the disagreement and say who holds which position.

Submission. By email to the panels address, with the moderator’s name and address included.

11.3 Case studies

Dates. “Deadline: September 22.” “Notification: October 6.”

What it is. “The Case Studies in Applied Security Track is a critical part of the technical conference. It is an opportunity for professionals to share information that is current without writing a detailed technical paper.” The track is “open to anyone in the community such as vendors, network providers, systems integrators, government civil/federal/military programs or users across the spectrum of computer security applications.”

What is submitted. Not a paper. “Submit a short paragraph describing your proposed presentation to the Case Studies Chair. This paragraph should indicate how the case study is important and relevant to today’s problems in computer security.” The submission carries the presenter’s name and affiliation, address, phone number, short biography, title, and abstract.

What gets published. “While the Case Studies will not be included in the Proceedings, the presentations will be posted to the ACSAC site following the conference.” A release is signed, and materials must not be marked restricted, proprietary, or company confidential.

What a good one looks like, quoted. It “Describes an application of security concepts or technologies used in a new way.” It “Describes a security architecture.” It “Displays a clear understanding of the relevance of this presentation to related efforts and products, including references as appropriate.” It “Is delivered without accompanying written material.” It “Is timely (current topics).”

What to avoid, quoted. “Presentations that could be used essentially unchanged as a sales or marketing pitches.” “Presentations that have been delivered before in a public forum, as a conference paper, magazine article, journal article, or book chapter.”

Award. “Yes! The committee will be awarding the ‘ACSAC Best Case Study Award’ at the conference.”

Obligation. Presenters “are required to register, pay, and attend the conference,” with reduced student registration and no exemption for organizers.

11.4 What the paper track hints say that the others do not

Two items from the Paper Hints and Notes page apply beyond the paper track and are worth having in one place.

The track redirection is official. “Papers which are not appropriate for the refereed papers track may be appropriate for the ‘case studies’ track. Another alternative is to assemble a group of people with diverse ideas on the topic you wish to discuss and submit a panel proposal.” The conference states the fallback route rather than leaving authors to infer it.

Anonymizing the artifact repository is the author’s job, and it is specified. “When preparing the artifacts repository for initial inclusion in the paper submission form . . . please take extra care to not include authors’ information in the repository or artifacts content, so as not to break the anonymity of the paper submission.” The page names services such as GitFront and Anonymous GitHub, and adds that “authors should make sure to use account names and repository names that do not identify the authors, and should remove any comments/text in the repository that may directly identify the authors or the authors’ institution.”

Two further items from the same page. Do not include a copyright notice, because it interferes with making copies for reviewers and with anonymization. And accepted submissions “may be assigned a member of the program committee to act as its shepherd through the preparation of the final

paper,” with that member acting “as a conduit for feedback from the committee to the authors.”

11.5 The three deadlines in order

Panels close first and by a wide margin. July 8 for the abstract, against September 19 for posters and September 22 for case studies. A panel is a decision made in spring, not in autumn.

11.6 Sources

- *Call for Posters and Works-in-Progress*, ACSAC 2026, acsac.org/2026/submissions/posters/. Read as a web page, 26 August 2026.
- *Call for Panels*, ACSAC 2026, acsac.org/2026/submissions/panels/. Read as a web page, 26 August 2026.
- *Call for Case Studies*, ACSAC 2026. Read in full from the local capture at `Submissions/2_Aug25th_May26th_ACSAC/Case Study.pdf`, text extracted with a converter.
- *Paper Hints and Notes*, ACSAC 2026. Read in full from the local capture in the same folder.
- Also in that folder and not used here: the main call for papers, *What Makes a Good Paper*, the artifact review guide, and the 2025 welcome and awards slides.

Verification status. The poster and panel pages were read once each, on one route. The case study and hints documents were read from local captures whose own capture date is not recorded in the file. Before relying on any date above, open the page again. Venue pages change quietly between years and these are dates rather than arguments.